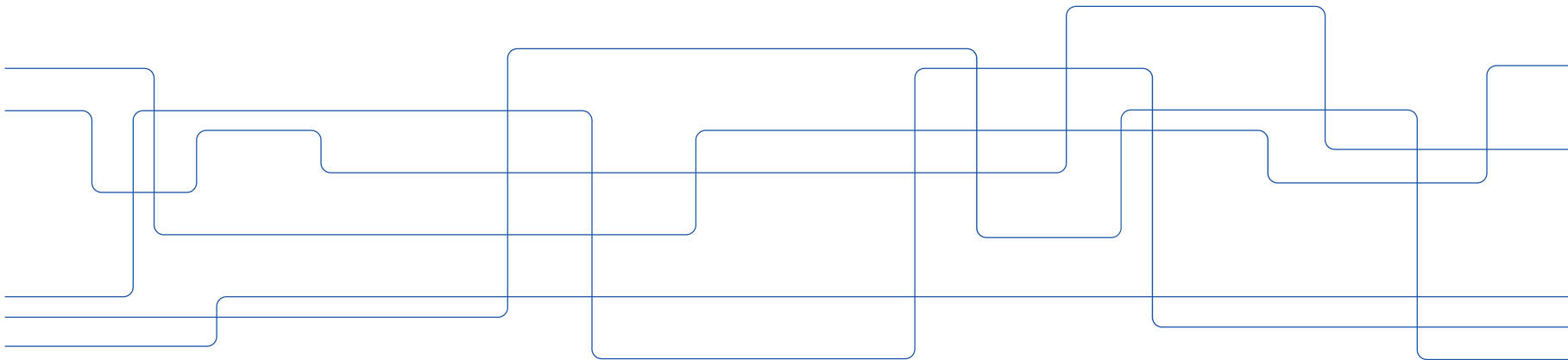


Neural Network Basics

(How deep learning is applied to supervised learning)



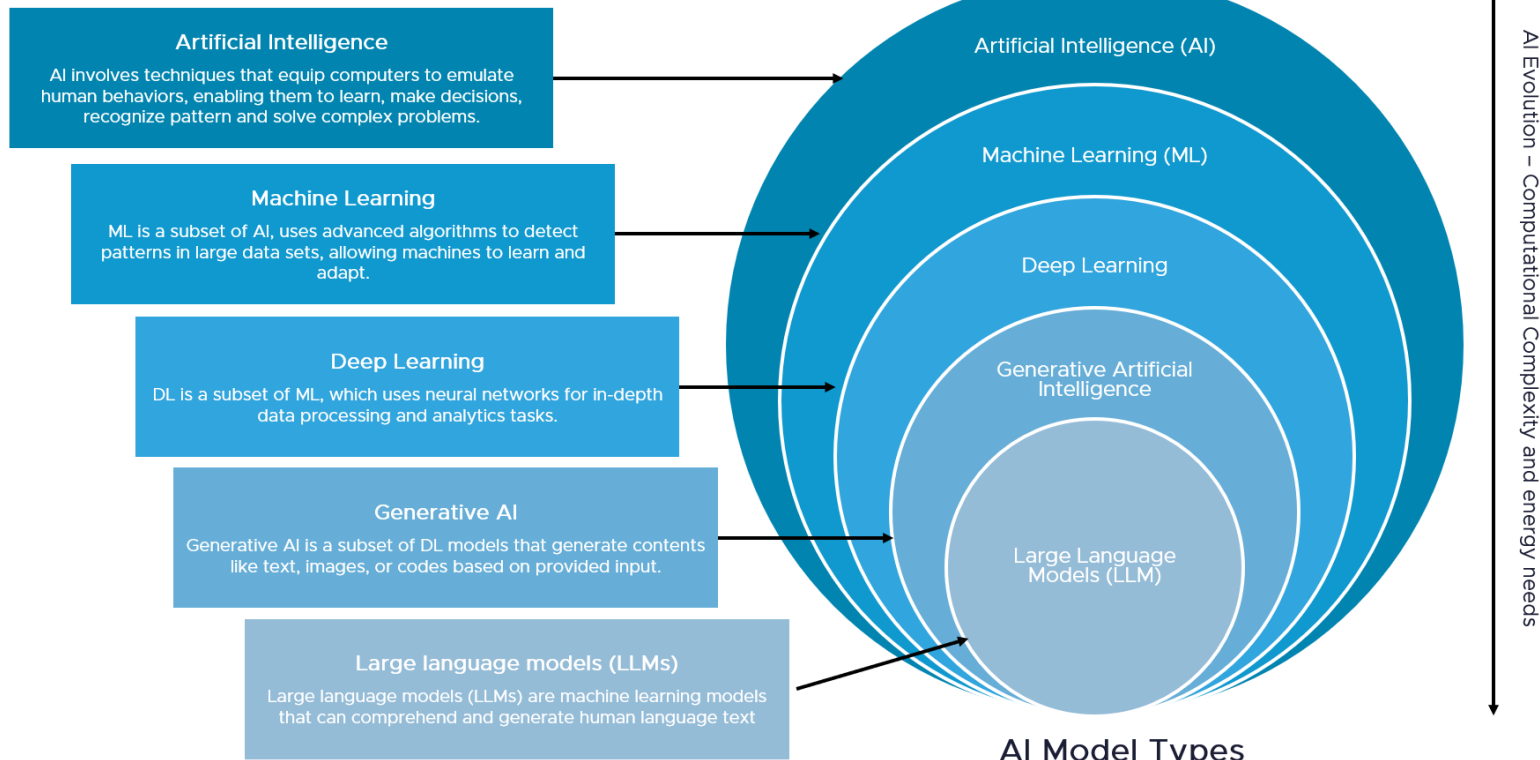
What we learn at the end of this session

- What is AI and ML?
- History of ML
- What is Neural Network?
- Logistic Regression as a simple Neural Network
- Quick Quiz Challenge



Introduction

AI Model Types

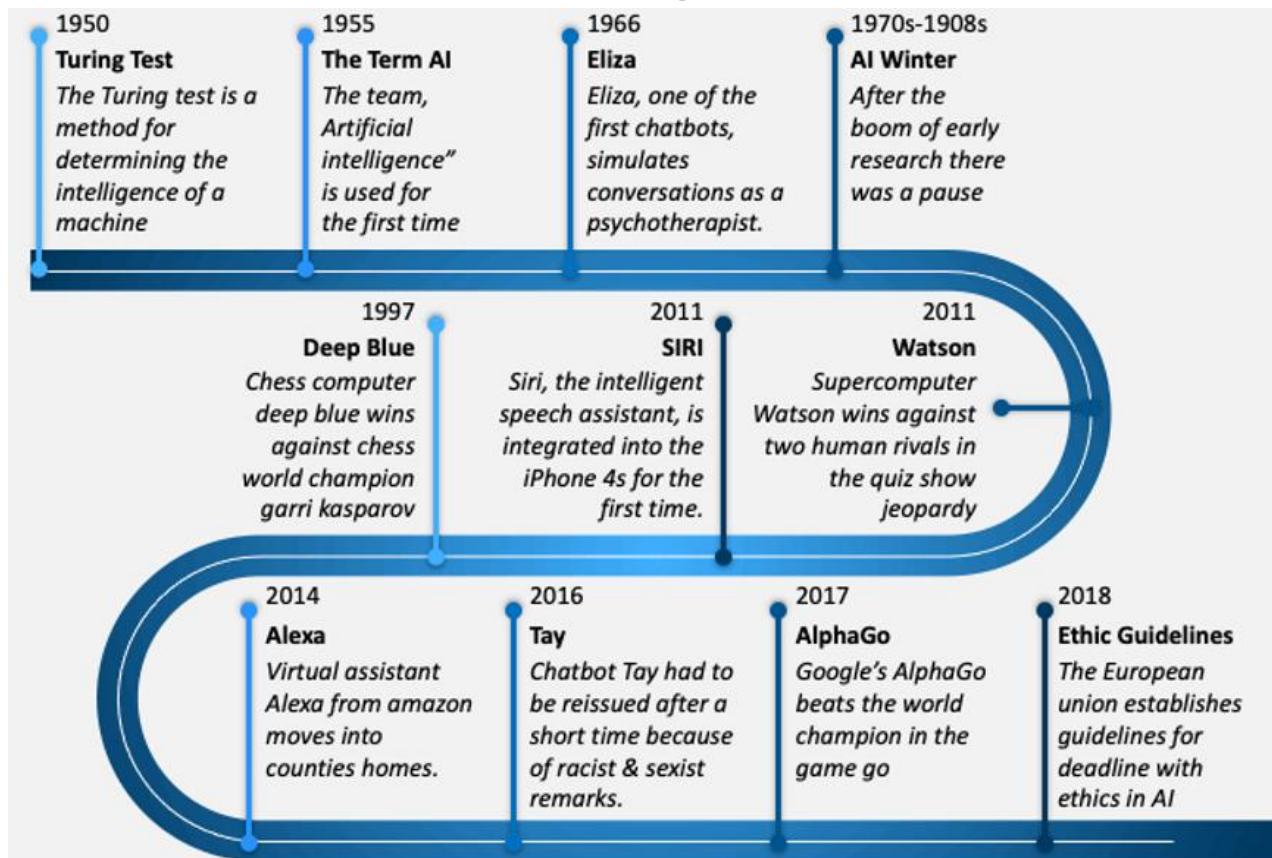




What is machine learning?

- For many problems, it's difficult to program the correct behavior **by hand**:
 - recognizing people and objects
 - understanding human speech
- Machine learning approach: program an algorithm to **automatically learn from data**, or from experience
- It's similar to **statistics**...
 - Both fields try to uncover patterns in data
 - Both fields draw heavily on calculus, probability, and linear algebra, and share many of the same core algorithms
- But it's **not** statistics!
 - Stats is more concerned with helping scientists and policymakers draw good **conclusions**; ML is more concerned with building **autonomous agents**
 - Stats puts more emphasis on **interpretability** and mathematical rigor; ML puts more emphasis on **predictive** performance, scalability, and autonomy

Introduction: history





Introduction: 2020-present

2020–2021: The Rise of Large-Scale Models

- GPT-3 Launch (2020): OpenAI released GPT-3, a massive language model with 175 billion parameters, showing advanced text generation and coding capabilities.
- Multimodal and Creative AI (2021): Early creative tools like DALL-E debuted, allowing systems to generate images from written prompts.

2022–2023: Mainstream Explosion

- ChatGPT Release (2022): Public access to ChatGPT brought generative AI into global mainstream use.
- GPT-4 and Competition (2023): More powerful multimodal models like GPT-4 and competitors like Anthropic's Claude emerged, handling text, images, and complex reasoning.

2024–2026: Regulation and Agentic AI

- Global Regulation (2024): The European Union passed the EU AI Act to establish legal frameworks for managing risks.
- Autonomous Agents (2025–2026): AI shifted from simple assistive chat co-pilots to autonomous agentic software capable of complex, multi-step workflows without human guidance.

Primary AI Methods

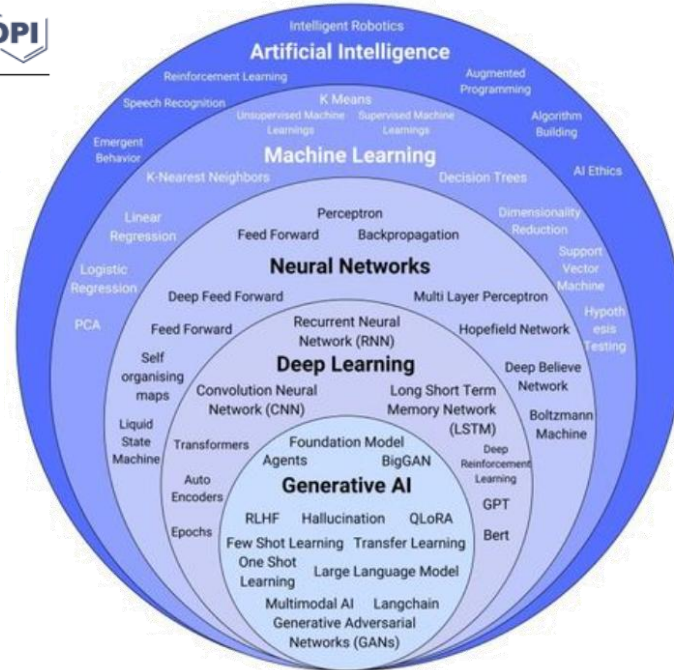
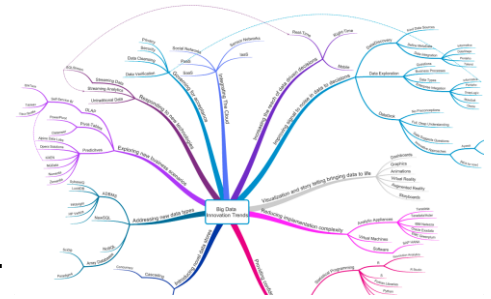
- There are a wide variety of methods, and still is under development.



Article

Synthesizing Multi-Layer Perceptron Network with Ant Lion Biogeography-Based Dragonfly Algorithm Evolutionary Strategy Invasive Weed and League Champion Optimization Hybrid Algorithms in Predicting Heating Load in Residential Buildings

- Regarding available data, complexity of the problem, computational computers, accuracy, simplicity, and etc the best method should be selected.





Types of machine learning

Supervised Learning:

Definition: Supervised learning is a type of machine learning where the algorithm learns from **labeled training data** to make predictions or decisions without human intervention.

Training Data: It requires a labeled dataset, meaning the input data is paired with the correct output or target.

Objective: The primary objective is to learn a mapping function from inputs to outputs, enabling the algorithm to predict the correct output for new, unseen inputs.

Examples: Common applications include fault detection, image classification, spam email detection, and regression tasks like predicting house prices.

Unsupervised Learning:

Definition: Unsupervised learning is a type of machine learning where the algorithm learns patterns and structures in data without explicit supervision. It aims to discover **hidden patterns**, group similar data, or reduce data dimensionality.

Training Data: Unsupervised learning uses **unlabeled data**, so there are no predefined target labels. It explores the inherent structure of the data.

Objective: The primary objectives include **clustering** data points into groups (clustering) or reducing the dimensionality of data (dimensionality reduction), making it easier to analyze.

Examples: Common applications include clustering customer segments for marketing, reducing the dimensionality of image data with techniques like Principal Component Analysis (PCA), and topic modeling in natural language processing.

Reinforcement Learning:

Definition: Reinforcement learning is a type of machine learning where an **agent learns** to make sequences of decisions by interacting with an environment to **maximize a cumulative reward signal**.

Training Data: Unlike supervised learning, reinforcement learning doesn't require labeled data. Instead, the agent learns from **trial and error** through interactions with the environment.

Objective: The primary objective is to find an optimal policy that defines the agent's actions in different states to maximize long-term rewards.

Examples: Reinforcement learning is used in autonomous robotics, game playing (e.g., AlphaGo), and recommendation systems.

Example of reinforcement learning

RL concept	Battery example
Agent	The AI/controller
Environment	Building + battery + electricity grid
State	Battery SOC, electricity price, PV generation, building demand, etc.
Action	Charge, discharge, or remain idle
Reward	Money saved, lower emissions, reduced peak demand, etc.

Electricity price = very low
AI decides → Charge battery
Later electricity price = very high
AI decides → Discharge battery
Result → saves €5
Reward = +5

But suppose the AI discharges when electricity is cheap:
Result → loses money
Reward = -2



Types of machine learning

With a focus on Neural Network (NN)

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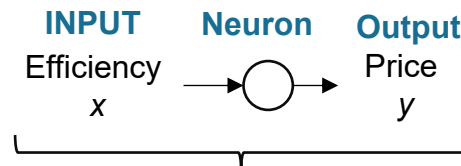
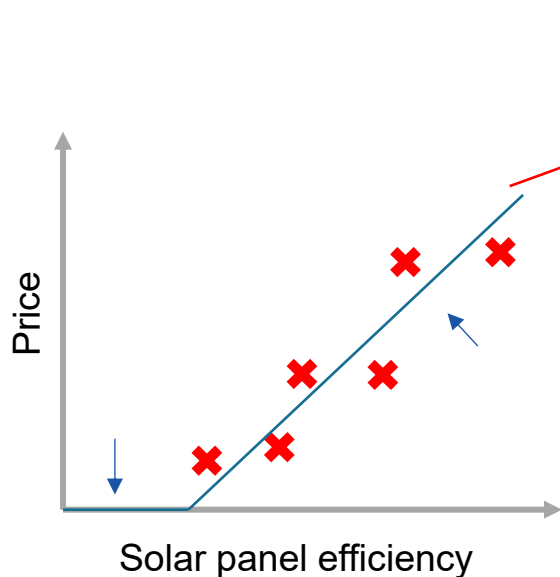
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Examples: Reinforcement learning is used in autonomous robotics, game playing (e.g., AlphaGo), and recommendation systems.

What is Neural Network?

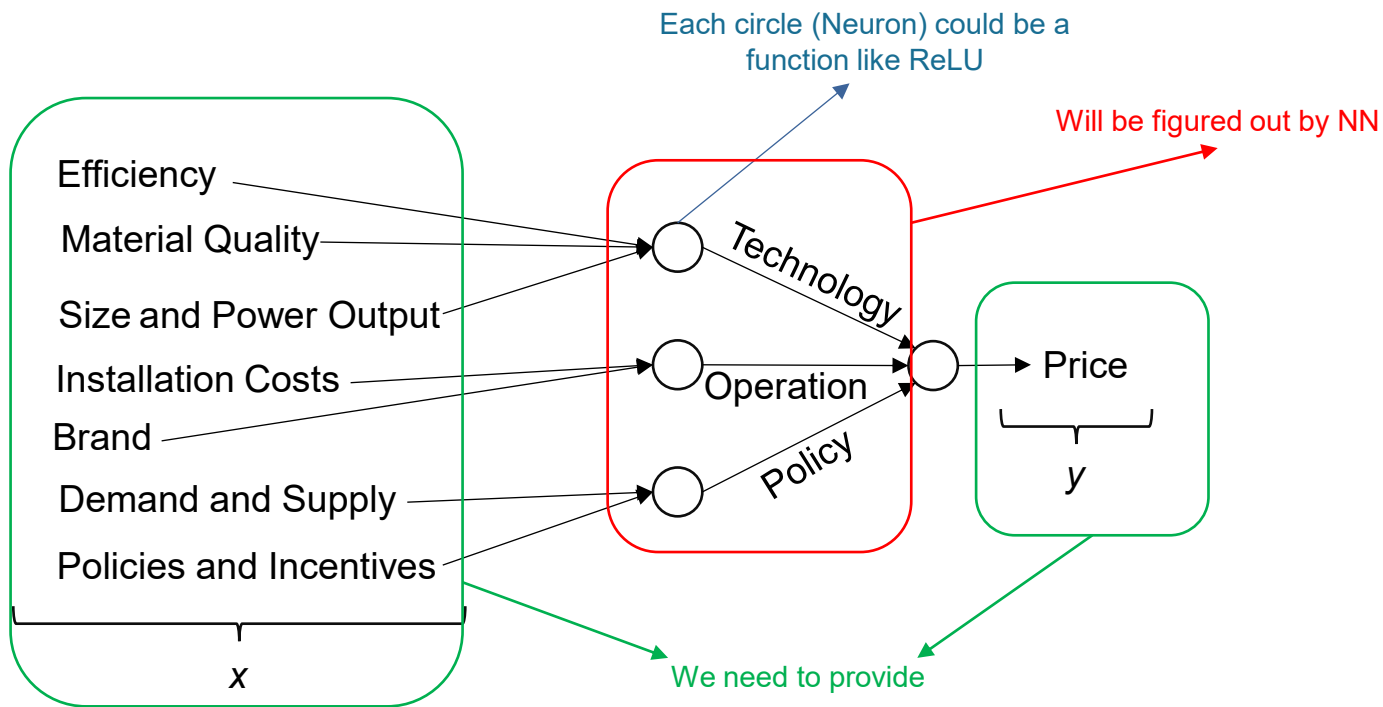
ReLU function
(Rectified Linear Unit)



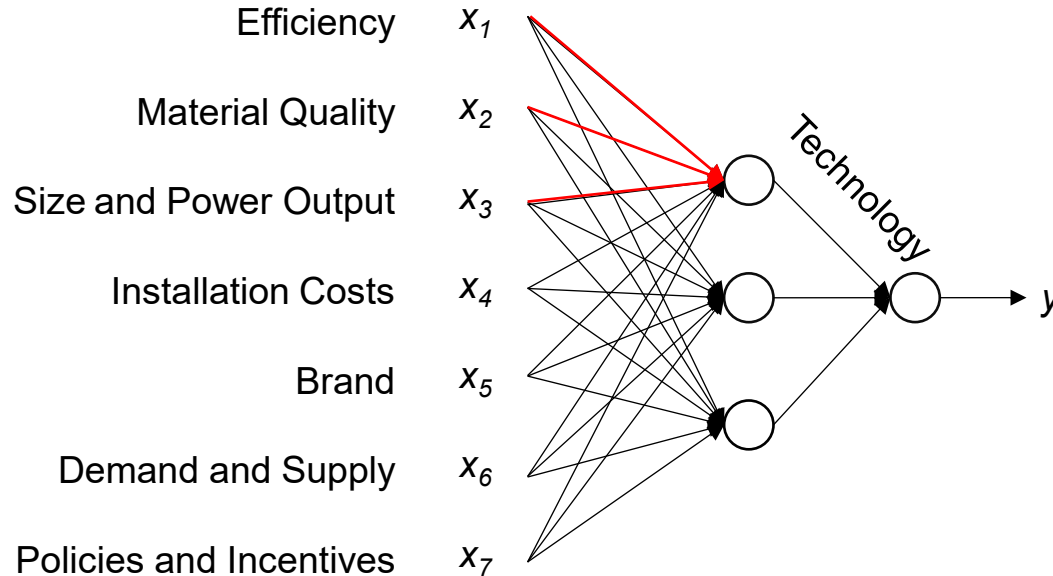
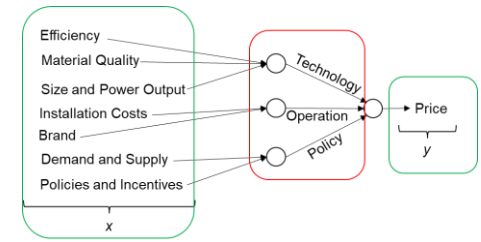
Single Neuron (Lego brick)

Get a bigger neural network by stacking together many of these Lego bricks

Solar panel price prediction



Solar panel price prediction



Input and Hidden layer are
densely connected

Notice also that each of these circles, these are called **hidden units in the neural network**, that each of them takes its inputs all seven input features. So for example, rather than saying the first node represents technology and technology depends only on the features x_1 , x_2 and x_3 . Instead, we're going to say, well neural network, you decide whatever you want this node to be.



Supervised learning with Neural Network

Applications

Input (x)	Output (y)	Application
Efficiency	Price	Energy Sector
Ad. user info	Click on ad? (0/1)	Online advertising
Image	Object (1,...,1000)	Photo tagging
Audio	Text transcript	Speech recognition
Swedish	Persian	Translation
Image, Radar info	Position of other cars	Autonomous driving

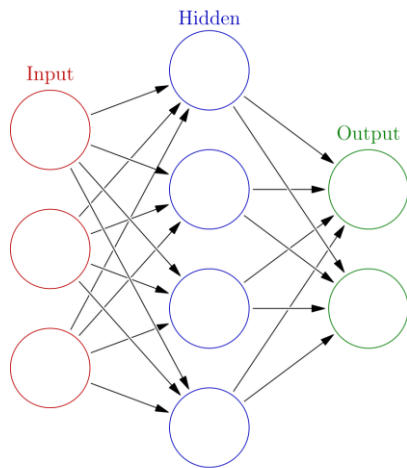
Standard NN

CNN

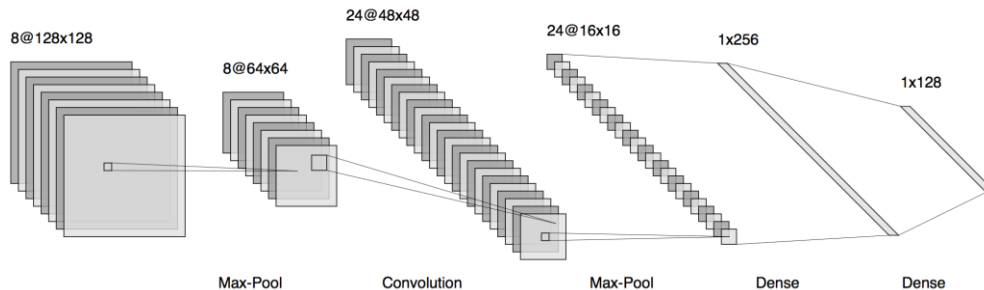
RNN

Custom/Hybrid

Neural Network examples

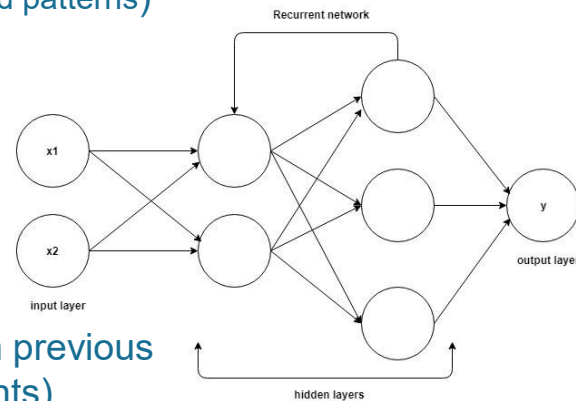


Standard NN



Convolutional NN

(to extract local features and patterns)



Recurrent NN

(utilize information from previous steps or time points)

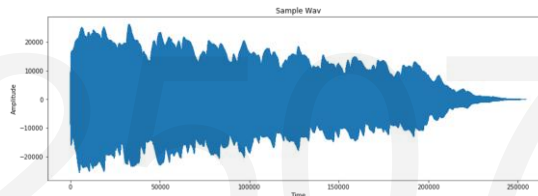
Supervised Learning

Structured Data

Efficiency (%)	Material	...	Price (\$/W)
15	silicon		0.3
20	silicon		0.8
22	Monocrystalline		1.0
18	Polycrystalline		0.7

User age	Ad ID	...	Click
41	93567		1
80	91927		0
27	46923		1

Unstructured Data



Audio

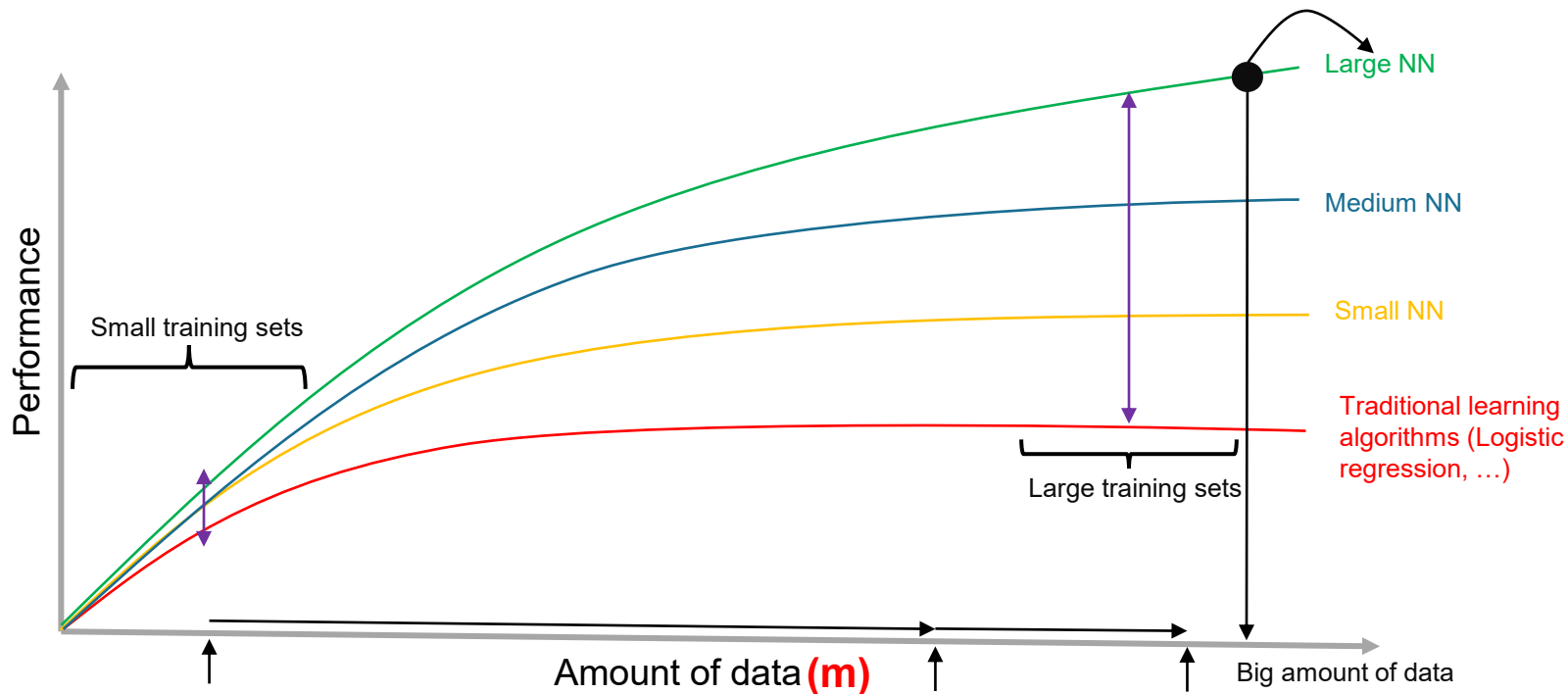


Image

It was the best of
times, it was the worst
of times, it was the age
of wisdom, it was the
age of foolishness...

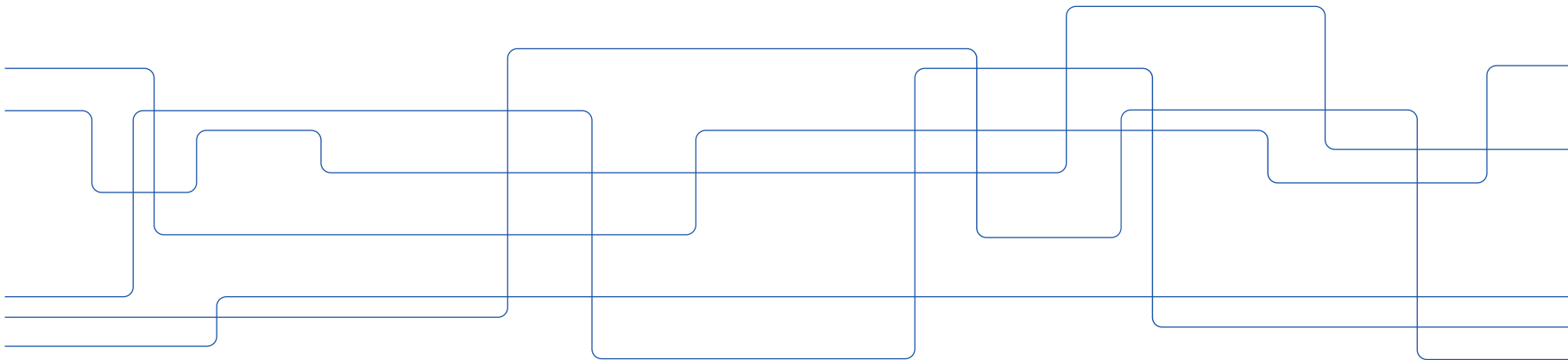
Text

Why has deep learning caught attentions?



Scale drives deep learning progress -> Scale: Size of NN, scale of data

Logistic Regression as a simple Neural Network (Build the general architecture of a learning algorithm, (gradient descent))





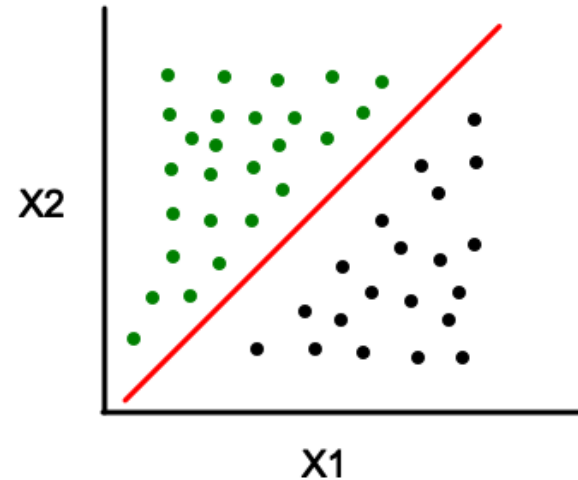
What Is Logistic Regression?

Logistic regression is:

- A supervised machine learning algorithm
- Accomplishes binary classification tasks by predicting the **probability of an outcome, event, or observation**

Key advantages of logistic regression:

- Easier to implement machine learning methods
- Suitable for linearly separable datasets
- Provides valuable insights



Notation

(x, y) x : Input parameters (a vector) y : output parameters

m training examples : $\{(x^{(1)}, y^{(1)}), (x^{(2)}, y^{(2)}), \dots, (x^{(m)}, y^{(m)})\}$

$(x^{(1)}, y^{(1)})$: training example 1, both $x^{(1)}$ and $y^{(1)}$ are matrix, meaning that they could be more than 1 value.

m_{train} = # of train examples and m_{test} = # of test examples

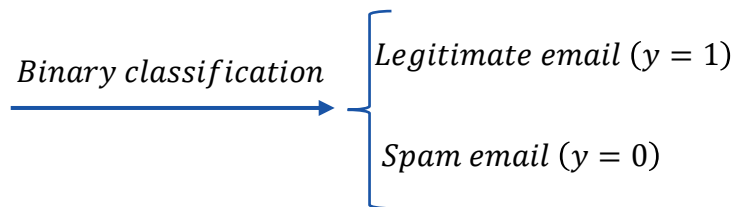
$$X = \begin{bmatrix} | & | & & | \\ x^{(1)} & x^{(2)} & \dots & x^{(m)} \\ | & | & & | \end{bmatrix} \begin{matrix} \updownarrow \\ n_x \\ \updownarrow \end{matrix}$$

$\xleftarrow{\quad m \quad}$

$$Y = [y^{(1)}, y^{(2)}, \dots, y^{(m)}]$$

$$Y \in \mathbb{R}^{1 \times m}$$

$$Y.shape = (1, m)$$



$$X \in \mathbb{R}^{n_x \times m} \quad X.shape = (n_x, m)$$



What Is Logistic Regression?

1. Start like Linear Regression

Suppose we want to predict whether a machine will **fail (1)** or **not fail (0)** based on temperature x .

We first calculate a linear score:

$$z = b + w_1x$$

For multiple inputs:

$$z = b + w_1x_1 + w_2x_2 + \cdots + w_nx_n$$

For example:

$$z = -5 + 0.1 \times \textit{Temperature}$$

If temperature = 60°C:

$$z = -5 + 0.1(60) = 1$$

The problem is that z can be anything:

$$-\infty < z < +\infty$$

But we want a **probability between 0 and 1**.

What Is Logistic Regression?

2. Pass it through the Sigmoid function

Logistic Regression uses the **sigmoid (logistic) function**:

$$P = \frac{1}{1 + e^{-z}}$$

This converts any number into something between **0** and **1**.

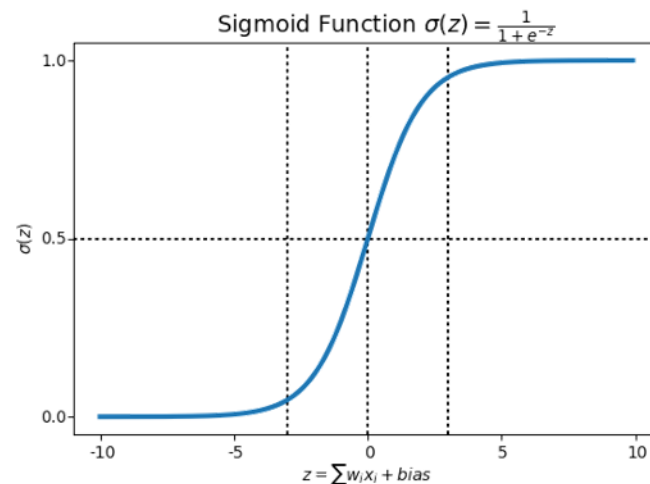
For our example, $z = 1$:

$$P = \frac{1}{1 + e^{-1}} \approx 0.73$$

So the model says:

Probability of machine failure = 73%

The sigmoid has an S-shape:





What Is Logistic Regression?

3. Convert probability into a class

We then choose a threshold, often **0.5**:

$$P > 0.5 \rightarrow \textit{Class 1}$$

$$P \leq 0.5 \rightarrow \textit{Class 0}$$

So if:

$$P = 0.73 \rightarrow \textbf{Failure}$$

and:

$$P = 0.20 \rightarrow \textbf{No failure}$$

But how does the model learn $b, w_1, \dots$?

During training, Logistic Regression searches for coefficients that make the observed classes as probable as possible.

For example, suppose the real result was **Failure = 1**:

Prediction 0.95 $\rightarrow$ very good

Prediction 0.80 $\rightarrow$ good

Prediction 0.10 $\rightarrow$ very bad

It measures these errors using a **cross-entropy loss function**, and an optimization algorithm adjusts the coefficients to minimize that loss.

Logistic Regression

Given X , we want $\hat{y} = P(y = 1|X)$

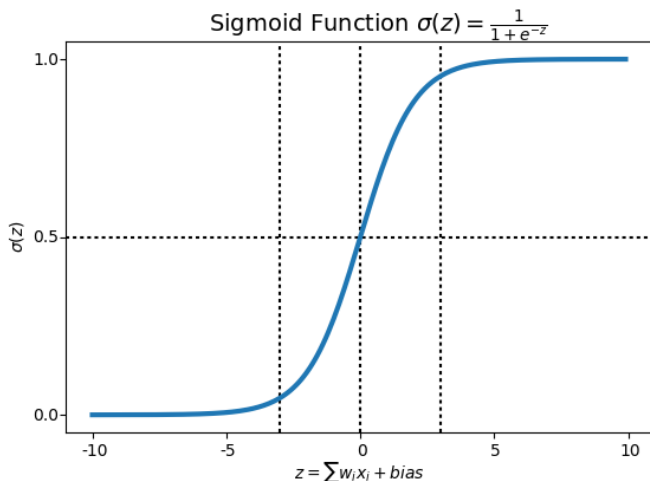
$$X \in \mathbb{R}^{n_x}$$

$$0 \leq \hat{y} \leq 1$$

Parameters : $\mathbf{w} \in \mathbb{R}^{n_x}$ and $\mathbf{b} \in \mathbb{R}$

Output: $\hat{y} = \sigma(\underbrace{\mathbf{w}^T X + \mathbf{b}}_z)$

So when you implement logistic regression, your job is to try to learn parameters $\mathbf{w}$ and $\mathbf{b}$ so that $\hat{y}$ becomes a good estimate of the chance of y being equal to 1.



$$\sigma(z) = \frac{1}{1+e^{-z}}$$

If z large, $\sigma(z) \approx \frac{1}{1+0} = 1$
 If z large negative number,

$$\sigma(z) = \frac{1}{1+\text{big number}} \approx 0$$

Logistic Regression cost function

$$\hat{y}^{(i)} = \sigma(w^T x^{(i)} + b), \quad \text{where } \sigma(z) = \frac{1}{1 + e^{-z}}$$

Given $\{(x^{(1)}, y^{(1)}), (x^{(2)}, y^{(2)}), \dots, (x^{(m)}, y^{(m)})\}$, want $\hat{y}^{(i)} \approx y^{(i)}$.

$$z^{(i)} = w^T x^{(i)} + b$$

$$\left. \begin{matrix} x^{(i)} \\ y^{(i)} \\ z^{(i)} \end{matrix} \right\} i^{th} \text{ example}$$

Loss function: apply on a single training example

$$L(\hat{y}, y) = -(y \log \hat{y} + (1 - y) \log(1 - \hat{y})) \rightarrow \text{We want to minimize this function}$$

$$\text{If } y = 1: L(\hat{y}, y) = -\log \hat{y} \rightarrow \text{want } \log \hat{y} \text{ large, want } \hat{y} \text{ large} \xrightarrow{\max(\hat{y})=1} \hat{y} = 1$$

$$\text{If } y = 0: L(\hat{y}, y) = -\log(1 - \hat{y}) \rightarrow \text{want } \log(1 - \hat{y}) \text{ large, want } 1 - \hat{y} \text{ large, want } \hat{y} \text{ small} \xrightarrow{\min(\hat{y})=0} \hat{y} = 0$$

Cost function: apply on a overall training example

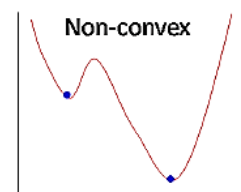
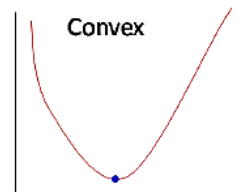
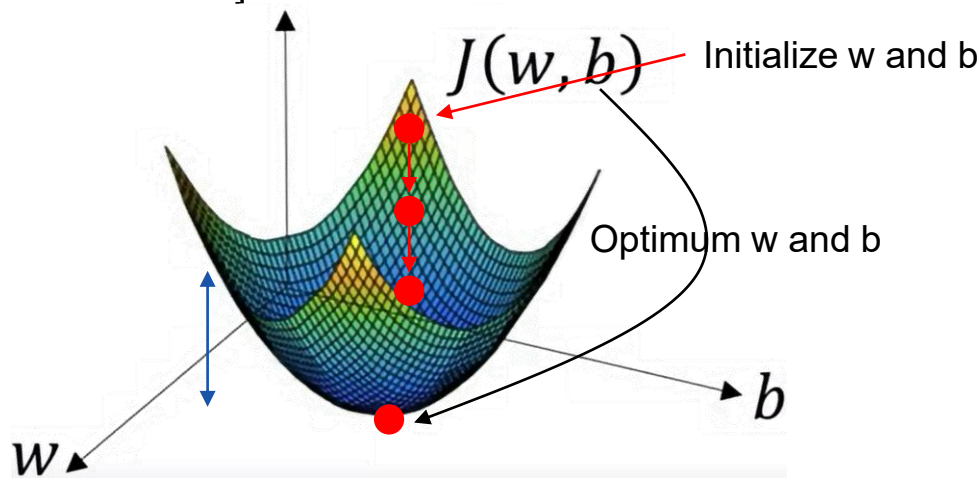
$$J(W, b) = \frac{1}{m} \sum_{i=1}^m L(\hat{y}^{(i)}, y^{(i)}) = -\frac{1}{m} \sum_{i=1}^m [y^{(i)} \log \hat{y}^{(i)} + (1 - y^{(i)}) \log(1 - \hat{y}^{(i)})]$$

Gradient Descent

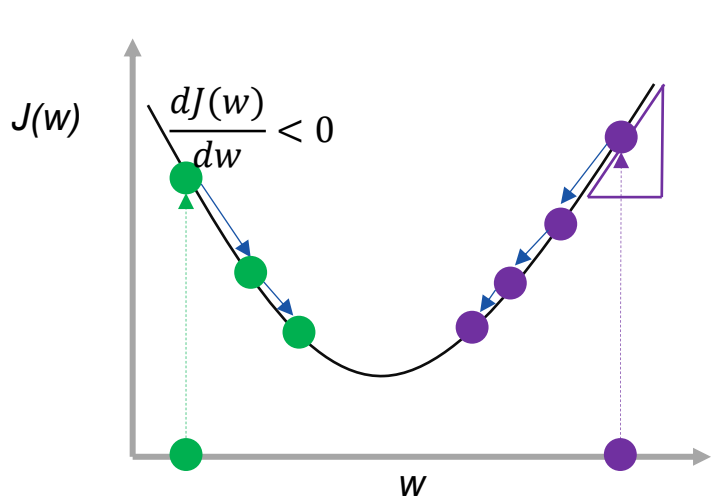
$$\hat{y} = \sigma(W^T X + b), \quad \text{where } \sigma(z) = \frac{1}{1 + e^{-z}}$$

$$J(W, b) = \frac{1}{m} \sum_{i=1}^m L(\hat{y}^{(i)}, y^{(i)}) = -\frac{1}{m} \sum_{i=1}^m [y^{(i)} \log \hat{y}^{(i)} + (1 - y^{(i)}) \log(1 - \hat{y}^{(i)})]$$

Want to find w, b that minimize $J(w, b)$



Gradient Descent



Repeat {

$$w := w - \alpha \frac{dJ(w)}{dw}$$

In codes: $w = w - \alpha dw$

$$J(w, b) \quad w = w - \alpha \frac{dJ(w, b)}{dw}$$

$$b = b - \alpha \frac{dJ(w, b)}{db}$$

$$\frac{\partial J(w, b)}{\partial w}$$

$$\frac{\partial J(w, b)}{\partial b}$$

Partial derivative
Functions of more
than 1 parameters

dw

db

Problem under study

Current condition



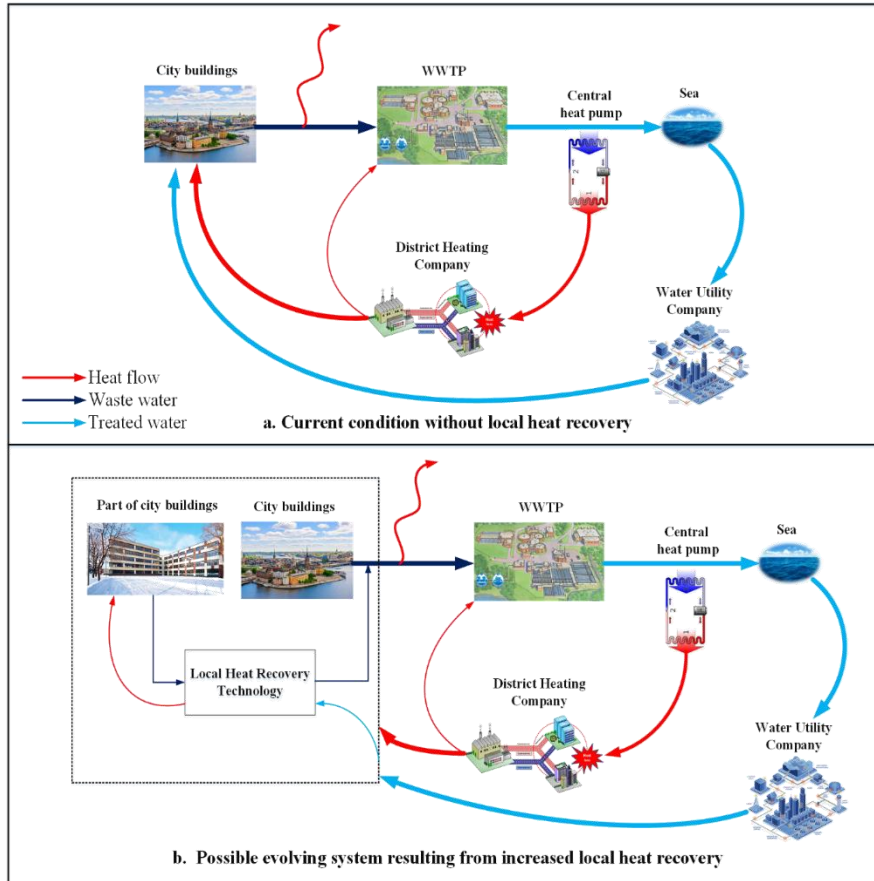
Stricter building regulations
&
Emerging new technologies



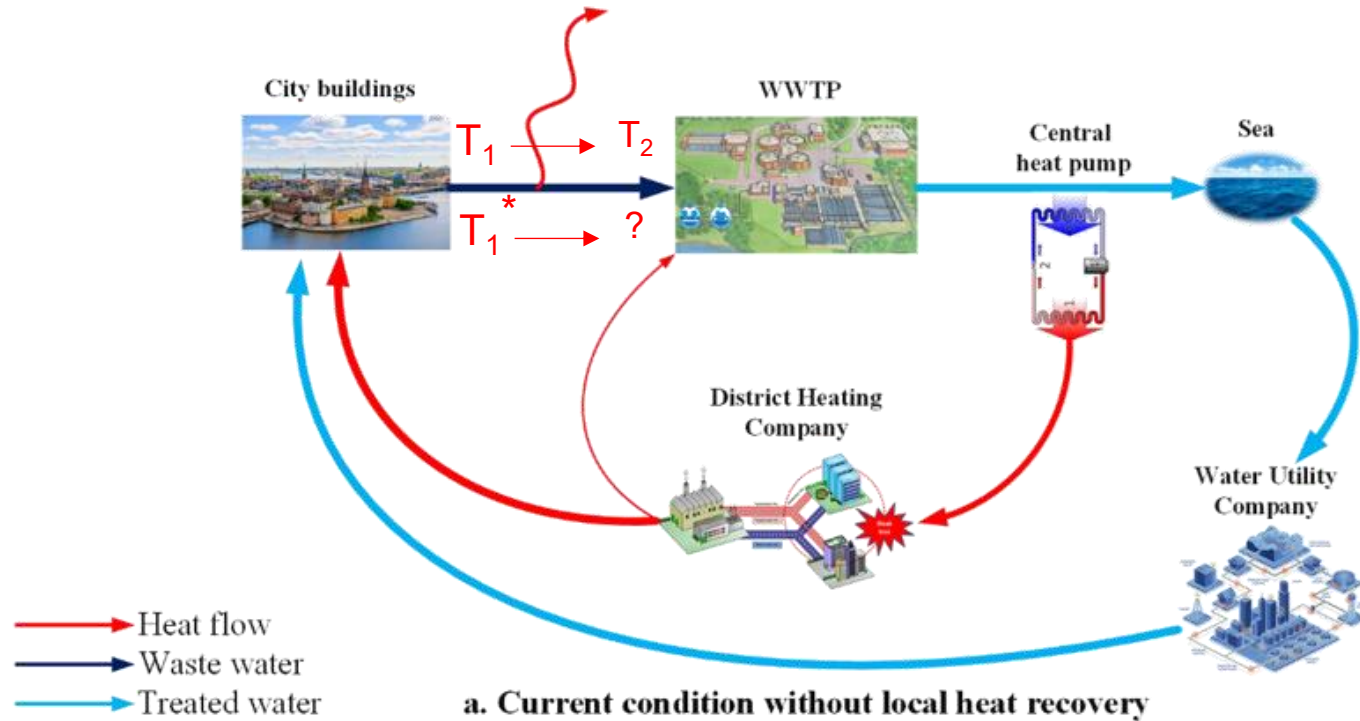
Local wastewater heat recovery



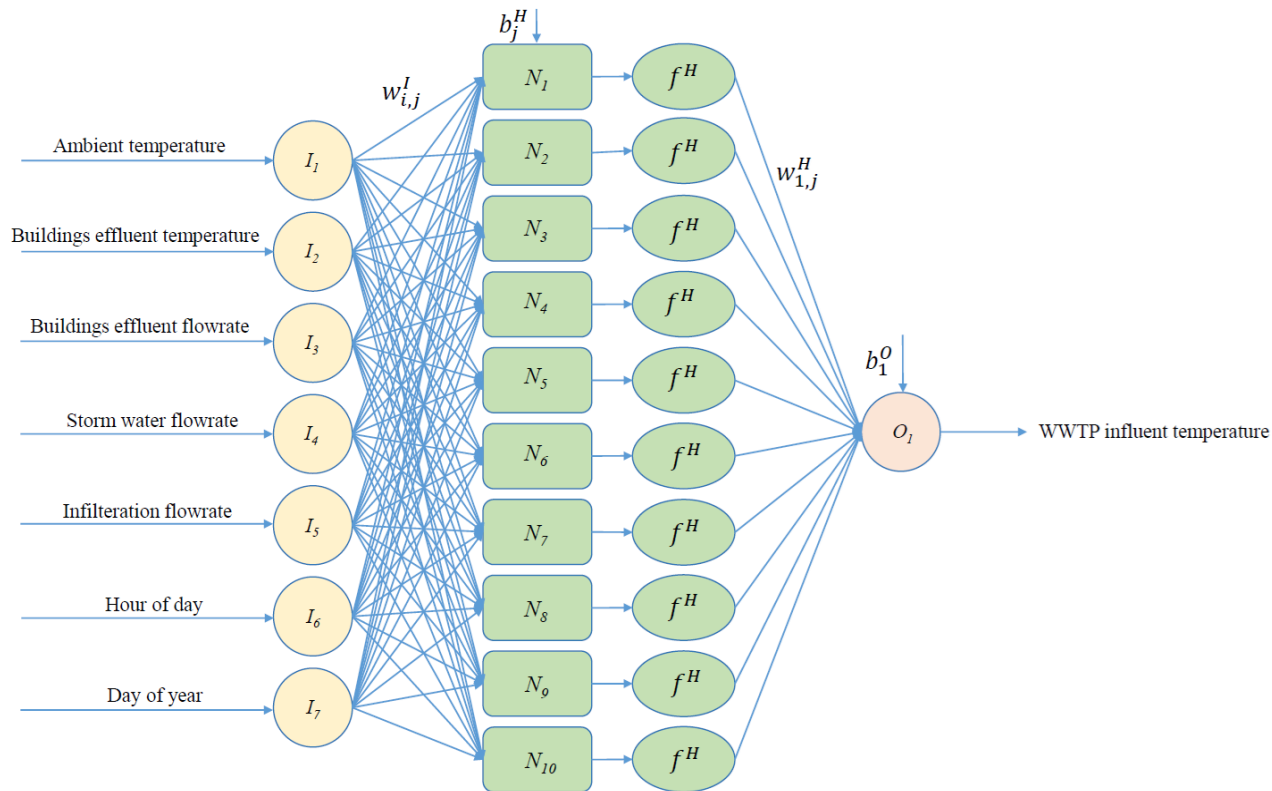
Possible future condition



Objective



Methods: Logistic regressions, Shallow NN, Deep NN





Quiz

Go to CANVAS and answer Quick Quiz Challenge 1



References

- [Speech and Language Processing](#). Daniel Jurafsky & James H. Martin. Draft of January 7, 2023.
- [Deeplearning.ai](#)
- [Implementing-a-neural-network-from-scratch/](#)
- [What Is Logistic Regression? Equation, Assumptions, Types, and Best Practices](#)